

THARUN RATHOD

AI Research · Machine Learning · LLM Evaluation

B.Tech Electrical Engineering · IIT Roorkee · 4th Year (2023–2027) · Roorkee, India

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PROFILE

IIT Roorkee EE student and independent AI researcher — five papers spanning LLM evaluation, behavioral AI, and applied ML, sole author on two, including a 484-trial study on LLM capitulation. Builds production AI systems solo, from MCP retrieval architectures to LLM-as-Judge pipelines to HMM/XGBoost forecasting. Seeking AI Research Internship roles.

EDUCATION

Indian Institute of Technology, Roorkee — B.Tech, Electrical Engineering

2023 – 2027

RESEARCH & PUBLICATIONS

The Capitulation Problem: A Conditional Optimization Framework

2026

- Sole Author, IIT Roorkee, 484 trials — Designed a behavioral experiment using Gemma 4B (Ollama) across 5 domains × 5 pressure variants with an LLM-as-Judge pipeline validated against blind human coding; found a 42.8% capitulation rate and the "Compliance Paradox" — direct challenge produced 64.6% hard reversals vs. 0% for expert-authority pressure.

Specialist, Generalist, and Hybrid LLM Architectures for Algorithmic Code Generation

2026

- Sole Author, IIT Roorkee, 75 problems + 4 probes ([github](#)) — Built a triaxial evaluation framework (Pass@1, CxSelf, CxJudge) across four architecture conditions; Generalist leads execution (0.911 vs. 0.862), Specialist leads complexity awareness, and a guarded Hybrid recovers 87% of a naive Hybrid's 0.133 regression.

Dual-Transformer Cross-Attention Multimodal Emotion Recognition

2026

- Co-Author, IIT Roorkee EE Dept — Led architectural decisions and authored the complete paper on a HuBERT + ViViT dual-encoder model with bidirectional cross-modal attention on the RAVDESS corpus, evaluated across 4 SNR robustness levels (15–0 dB).

Physics-Guided Diffusion Models for Synthetic Power Flow Data Generation

2025

- Co-Author, IIT Roorkee EE Dept — Contributed to a physics-guided DDPM framework enforcing AC power flow feasibility, achieving a 91.4% feasible fraction vs. a 41.2% unconstrained baseline on PJM 5-bus and IEEE bus systems.

Phonological Fidelity and Convergent Preservation in Indo-European Languages

2026

- Sole Author, Open Access ([Zenodo](#)) — Developed the Phonological Fidelity Index (PFI), a quantitative measure of consonant retention from Proto-Indo-European roots across French, Hindi/Sanskrit, and English under Grimm's Law.

AI / ML PROJECTS

NIFTY-50 – Investment Intelligence Platform

2026

- Solo, Python/hmmlearn/XGBoost/SHAP ([Live Demo](#)) — Built a Hidden Markov Model to classify market regimes (Bull/Bear/Sideways), detecting shifts 3–5 days early, and layered XGBoost + SHAP price prediction with Sharpe/Sortino/VaR risk analytics on a live dashboard.

IITR Nexus – Unified Campus Intelligence Platform

2026

- Solo, Next.js/TypeScript/Groq Llama 3.3 70B/MCP ([live](#)) — Architected a multi-MCP router that classifies intent and dispatches a query in parallel across six domain connectors; below a confidence threshold it returns a source reference instead of generating an answer.

EXPERIENCE

R.K. IntelliServe · Machine Learning Trainee · Gurugram, Haryana

Dec 2025 – Feb 2026

- Engineered and automated DAG-orchestrated ML preprocessing and evaluation pipelines for production data-processing workflows.

Hiring Bazaar · Web Developer Intern · IIT Roorkee Startup

Oct 2025 – Jan 2026

- Built the careers page, dynamic role listings, and candidate application workflow across the frontend and backend, and owned UI feedback and testing during rollout.

TECHNICAL SKILLS

Languages

Python, TypeScript, JavaScript, C++, SQL

AI / ML

LLM-as-Judge Eval, Prompt Engineering, MCP Servers, RAG, HMM, XGBoost, SHAP, scikit-learn, Ollama, Groq, Gemini, Claude API

Research & Systems

Experiment Design, Statistical Analysis, Human-Coding Validation, LaTeX, Node.js, FastAPI, PostgreSQL, MongoDB, Docker, Git

CERTIFICATIONS & ACHIEVEMENTS

MongoDB Core Concepts & Architecture (Jun 2026) · AI Product Management, Simplilearn (Jun 2026) · AIR 23,496 – JEE Adv. 2023 (top ~1%) · EAPCET Rank 2,614 · NCC Cadet, IIT Roorkee · English, Telugu, Hindi